CSC 352 - Systems Programming and UNIX
Fall 2025

TuTh 2:00PM-3:15PM, Ctr English 2nd Lng, Rm 103

Course Description

Programming in C, including arrays, lists, stacks, queues, trees, pointers, and dynamic memory allocation.

Unix topics, including debuggers, makefiles, shell programming, and other topics that support C programming.

Catalog Description:
Programming in C, including single and multi-dimensional arrays, lists, stacks, queues, trees, and bit manipulation. Unix topics, including debuggers, makefiles, shell programming, and other topics that support systems programming.

Instructor and Contact Information

Instructor:

Eric Anson

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TAs:

Talha Abrar	tabrar@arizona.edu
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Office Hours

(Exact office hours subject to change, consult piazza for up-to-date information.)

Additional hours may be available by appointment.

Eric Anson

Mon, 1:00-2:30, GS 809

Tue, 3:30 – 5:00, GS 809

TAs Office Hours

TBA

Websites

Piazza: https://piazza.com/arizona/fall2025/csc352/home

D2L will be used for distributing grades, posting videos, and quizzes.

GradeScope will be used for returning exams and quizzes.

Course Format and Teaching Methods

There will be two lectures a week. Attendance for the lectures is mandatory. Out-of-class activities will include reading about and exploring interacting with Unix and C as well as a significant number of programming projects.

Course Objectives

- Learn about using the Unix operating system and the bash shell.
- Learn the C programming language including memory allocation and management and the use of pointers.
- Learn various programming tools available in the Unix environment.
- Get further experience in designing, writing, testing, and debugging code.

Expected Learning Outcomes

Students will be able to:

- Being able to interact with a Unix environment via a Bash shell including navigating the file system, being able to copy, delete, edit, and search files, understanding and utilizing I/O redirection and pipes
- Understanding of, and facility with, Unix program development tools such as **make**, **gcc**, **valgrind**, etc.
- Being able to understand, write and debug simple Bash shell scripts.
- Being able to understand, write and debug complex C programs that involve multiple source files, header files, pointer manipulation, and dynamic memory management.

Transferable Career Skills

National Association of Colleges and Employers (NACE) Career Readiness:

Career readiness is a foundation from which to demonstrate requisite core competencies that broadly prepare the college-educated for success in the workplace and lifelong career management. For new college graduates, career readiness is key to ensuring successful entrance into the workforce.

There are eight career readiness competencies, each of which can be demonstrated in a variety of ways." (NACE, 2025)

- Career & Self Development
- Communication
- Critical Thinking
- Equity & Inclusion

- Leadership
- Professionalism
- Teamwork
- Technology

In this course, we will focus on the following competencies:

- Technology: Students will learn the C programming language which is commonly used in operating systems, compilers, and other system software as well as in the game industry and in embedded systems. Students will also learn Unix commands and command line. Unix is the most common OS for servers and is also used for embedded systems, switches and routers, robotics, and personal computers.
- Critical Thinking: In this class we will be writing many fair-sized programs. Being a good programmer requires the ability to think critically and logically about the best way to solve problems.
- Professionalism: Completing the assignments on time will require good time management. This is critical skill in the professional world.
- Communication: Students are encouraged to emphasize communication by interacting with teaching assistants and using various channels, such as D2L, email, class discussions, and piazza to stay updated on course materials.

Makeup Policy for Students Who Register Late

Students who register late will not be able to make up the missed work.

Course Communications

The primary path for outside-lecture questions will be Piazza. The instructor and TAs can also be reached through email. Assignments, announcements, and documents pertaining to the class will be posted on D2L. All code and executables will be made available through a shared directory on the department computer called lectura.

Required Texts or Readings

- Stephen Prata, *C Primer Plus*
- Cameron Newham and Bill Rosenblatt, *Learning the bash Shell*, 2nd Edition. O'Reilly Media, Inc. Print ISBN-13: 978-1-56592-347-8.

There are many good books on C programming, Unix, and bash programming. The student is encouraged to explore any of these.

Assignments and Examinations: Schedule/Due Dates

<u>Projects</u>

This class will include several programming projects. The following schedule gives approximate due dates for each one. The schedule below is approximate; as we assign the projects, it may occasionally be necessary to adjust due dates. Likewise, the topics mentioned are our current plans are subject to change.

It is possible that we might assign **fewer** projects than listed below; however, we will not assign more.

Project 1	due on or about Fri, Sep 12	Introduction to C
Project 2	due on or about Fri, Sep 19	Basic C constructs, Bash scripting

Project 3	due on or about Fri, Sep 26	First experience with arrays and/or pointers
Project 4	due on or about Fri, Oct 3	More experience with pointers
Project 5	due on or about Fri, Oct 10	Introduction to malloc()
Project 6	due on or about Fri, Oct 24	Practice with malloc()
Project 7	due on or about Fri, Oct 31	Introduction to free()
Project 8	due on or about Fri, Nov 7	Practice with free()
Project 9	due on or about Fri, Nov 21	Misc topics in C; also continue w/ pointers
Project 10	due on or about Fri, Dec 5	Misc topics in C; also continue w/ pointers
Project 11	due on or about Wed, Dec 10	Misc topics in C; also continue w/ pointers

Projects will be due at 11:59 pm; most or all projects will be turned in electronically, using the "turnin" program on Lectura. There will be an 8 hour "grace period" during which projects will continue to be accepted without penalty, but after that late work will not be accepted. Additionally, this class does not allow re-submission of work after the due date.

Midterms

We will have two midterm exams. They will be held during the regular lecture times in our regular classroom. They are tentatively scheduled for Thu 10/16 and Thu 11/13.

Pop Quizzes

Pop quizzes will be given frequently. These quizzes will be given in class. Students must be present to take the quizzes. No makeup quizzes will be given, but the lowest 1/5 of the quiz grades (in favor of the student) will be dropped to allow for unavoidable absences.

Written Assignments

There is the possibility of a few short, written assignments. Some of these assignments might be online.

Final

There will be a cumulative final which will be given in our regular classroom on:

Mon, Dec 15, 3:30 – 5:30 PM

Final Examination

There will be a cumulative final given.
The final is scheduled for
Mon, Dec 15, 3:30 – 5:30 PM

https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams

Grading Scale and Policies

Grading Scale

We will use a simple grade cutoff scheme. This means that if you earn the number of points listed for a given grade, you are guaranteed that grade. At the end of the semester, we reserve the right to **lower** these cutoffs (meaning that it might be easier to earn a good grade); however, we do not guarantee that we will do so. However, we guarantee that we will not raise these cutoffs (making it harder to earn a good grade).

- 90% A
- 80% B
- 70% C
- 60% D

<u>Point Distribution</u>

Points will be distributed as follows:

- 10% Quizzes & Written Assignments
- 40% Programming Assignments
- 30% Midterms
- 20% Final Exam

Within each category, points are distributed evenly; that is, every quiz is worth the same as every other, and every assignment is worth the same as every other. Thus, the exact value of each quiz or assignment will depend on the number of each event that is assigned.

<u>Grading Schedule</u>

Projects will be graded, typically, within 6 days of the due date. If exceptions must be made occasionally, staff will inform the students about the delay and the reason for it.

Pop quizzes and exams will be graded within 10 days.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at https://catalog.arizona.edu/policy/courses-credit/grading/grading-system.

Dispute of Grade Policy If you have an issue with how a quiz, project, or exam was graded, you must submit a request for a regrade (through email) to within 7 days of when the item was returned to you.

Scheduled Topic and Activities

This is a list of the topics we will cover. The dates may change slightly. Every semester the pace moves a little differently based on student questions, etc. There are no assigned readings for these topics. While outside reading in the recommended (or other) texts is encouraged, all information required will be on the slides which will be posted on D2L.

Week 1	Aug 26, 28	Introduction, Basic Unix
Week 2	Sep 2, 5	Basic Unix, Basic C
Week 3	Sep 9, 11	Basic C, basic shell scripting
Week 4	Sep 16, 17	Arrays, Structs, Pointers
Week 5	Sep 23, 25	Arrays, Structs, Pointers, debuggers
Week 6	Sep 30, Oct 2	debuggers, basic make
Weed 7	Oct 7, 9	cmd line arguments, file IO
Week 8	Oct 14, 16	free, Exam 1
Week 9	Oct 21, 23	the C preprocessor, separate compilation
Week 10	Oct 28, 30	make, code coverage
Week 11	Nov 4, 6	separate compilation, make
Week 12	Nov 11, 13	Veterans Day!, Exam 2

Week 13	Nov 18, 20	system calls, signal handling
Week 14	Nov 25, 27	program hygiene, Thanksgiving!
Week 15	Dec 2, 4	fork, more Unix
Week 16	Dec 9, 11	more Unix, performance tuning

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

Some learning styles are best served by using personal electronics, such as laptops and iPads. If these are used for other purposes (games, web browsing, videos, etc) they can be very distracting for other students. Please only use devices for classroom related activities during the lectures. In this class bringing laptops is encouraged as students might want to try bits of code during the lectures.

Academic Integrity

The work done in this class must be your own. Copying code from any source other than work you have written prior on your own is forbidden. Using an AI to help you generate code is forbidden. Sharing or showing your code with other students is forbidden.

Prior to the widespread use of AI I almost never had any academic integrity violations in this class. The last few times I have taught this class there have been several. Even more common than the students caught, I have had students who do extremely well on the projects using advanced methods that we haven't covered in class, that then fail the exams and due very poorly in the class. The goal of this class is to learn Unix and C, not how to feed problems to a chat bot. I don't know what AI will mean for the future of programming jobs, but if you want stay ahead you need to know more than the AI.

Academic integrity violations will be treated in a case by case basis. The MINIMUM punishment will be loss of all points on the assignment in question. The maximum punishment will be a drop in letter grade. These results are per case. Also remember that all academic integrity violation incidents have to be reported to the dean who might add further repercussions.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): https://cirt.arizona.edu/case-emergency/overview

Also watch the video available at

https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/app/me/ledetail;spf-url=common%2Flearningeventdetail%2Fcrtfy000000000003841

University-wide Policies link

Links to the following UA policies are provided here: https://catalog.arizona.edu/syllabus-policies

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy
- Class Recordings
- Additional Resources
- Preferred Names and Pronouns

Department-wide Syllabus Policies and Resources link

Links to the following departmental syllabus policies and resources are provided here, https://www.cs.arizona.edu/cs-course-syllabus-policies :

- Department Code of Conduct
- Illnesses and Emergencies
- Obtaining Help
- Confidentiality of Student Records
- Land Acknowledgement Statement

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.